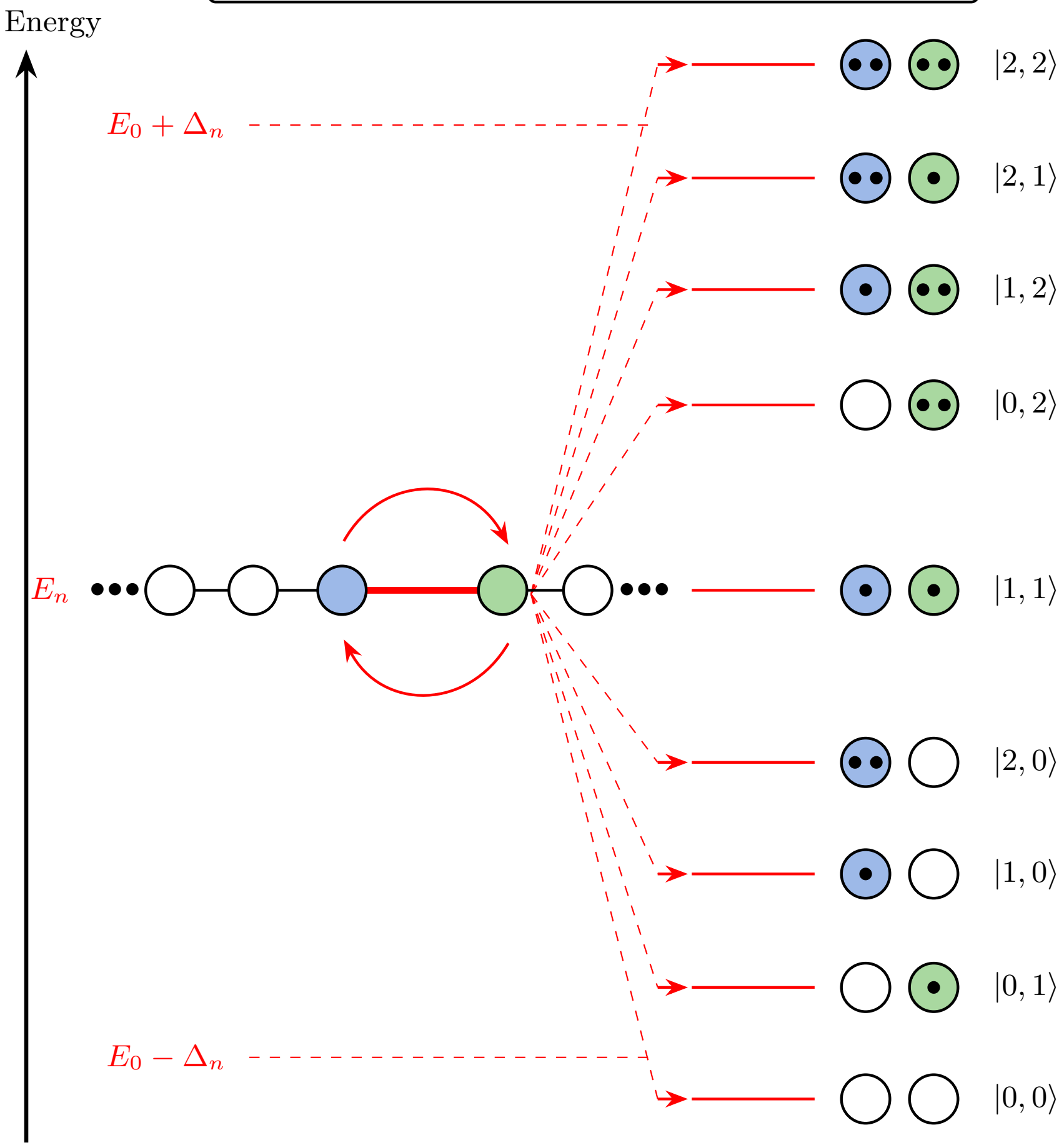

(A) Coupling of $|1,1\rangle$ to virtual (off–resonant) states
Energy
$E_0 + \Delta_n$
E_n
$E_0 - \Delta_n$
$|2,2\rangle$
$|2,1\rangle$
$|1,2\rangle$
$|0,2\rangle$
$|1,1\rangle$
$|2,0\rangle$
$|1,0\rangle$
$|0,1\rangle$
$|0,0\rangle$